INTERNAL ENGINEERING BRIEF

Call Quality Improvement Plan

Raising Talkly's voice-call quality: speech recognition, voice output, and conversational reasoning.

Prepared for Novalantis / TalklyAI engineering • 27 October 2026

Talkly's live voice pipeline (Plivo ↔ OpenAI Realtime) now runs with sub-500ms response latency after the pre-warming fix, and recent transcription-model and prompt-rule changes have closed the worst language-mixing and premature-hangup bugs. Three quality gaps remain, all traced to the same root cause: **OpenAI's Realtime API bundles speech recognition, reasoning, and voice output into a single locked pipeline**, which caps what any prompt or config tweak can fix. This brief diagnoses each gap with evidence from real test calls and lays out a phased path forward — starting with a zero-risk model swap, through to the architecture decision required for an authentic Indian-accented voice.

Speech Recognition	Voice Output	Reasoning
Hallucinates on silence	No Indian-accented option	Mini model skips own rules

1. Current Architecture Snapshot

Every live call runs through a single WebSocket bridge connecting Plivo's telephony stream directly to OpenAI's Realtime API — one model handles speech-in, conversational reasoning, and speech-out together.

Layer	Current Choice	Constraint
Telephony	Plivo (direct REST dispatch)	Fixed
Speech-to-Text	whisper-1 (via Realtime session)	Shares OpenAI's silence-hallucination behavior
Reasoning / “Brain”	gpt-realtime-mini	Weaker multi-rule instruction adherence than full model
Text-to-Speech	OpenAI Realtime voice: “coral”	1 of 10 fixed voices — none Indian-accented

Why this matters: because all three layers live inside one OpenAI-hosted session, no individual layer can be replaced without breaking apart the pipeline into a cascaded STT → LLM → TTS design run across separate vendors. That single architectural fact is the thread running through every finding below.

2. Diagnosis — Three Failure Modes, With Evidence

2.1 Speech Recognition: hallucinates fluent text on silence

Whisper-family models (whisper-1, and the gpt-4o-transcribe family it would be swapped for) are documented to fabricate coherent stock phrases — sign-offs, thank-yous, generic blessings — when fed near-silent or very low-energy audio, rather than returning nothing. This is a training-time property of the model family, not a bug reachable by prompt or VAD tuning.

```
08:03:32.968  VAD: customer started speaking
08:03:36.349  VAD: customer stopped speaking
08:03:36.873  CUSTOMER SPEECH (transcribed): "I love you all. God bless you all."
    → in response to: "Which day and time works best for the demo?"
```

Real excerpt, call quick-call-1793088114280, 27 Oct 2026 — the agent then invented its own "Saturday 11 AM" slot rather than re-asking.

2.2 Voice Output: no Indian-accented option exists in the current vendor

OpenAI Realtime ships exactly 10 fixed voices (alloy, ash, ballad, coral, echo, sage, shimmer, verse, marin, cedar) — all Western-accented. No prompt instruction can change a voice's underlying accent; the "speak with an Indian accent" directive already in the system prompt can only nudge pacing and word choice, not phonetics. This is a vendor catalogue limit, not a prompting gap.

2.3 Reasoning: the mini model inconsistently follows its own rules

The live conversational model is **gpt-realtime-mini**, operating under a dense system prompt (accent directive + 6 conversation rules + staged objective flow). In the same call above, rule #6 ("if the input doesn't logically fit, ask the customer to repeat") did not fire on the hallucinated transcript — the model proceeded to invent a plausible-sounding appointment slot instead. Mini models are measurably less reliable at consistently obeying long, multi-rule instruction sets than their full-sized counterparts; this is a known general property of distilled models, not specific to this prompt.

3. Recommendation Matrix

Area	Option	Fixes	Latency Impact	Architecture Change
Brain	**gpt-realtime (full)**	Rule adherence, objection handling	Small increase	None — model string swap
STT	**Deepgram Nova-3**	Silence hallucination (per-word confidence scoring), 8kHz telephony tuning	Pipeline moves from ~400ms to ~800ms–1.2s	Breaks Realtime bundle — cascaded pipeline
TTS	**Rime AI / Sarvam AI**	Authentic Indian-English / Hindi voice	~200ms TTFB (Rime) — comparable	Breaks Realtime bundle — cascaded pipeline

STT and TTS land in the same column for a reason: both require leaving OpenAI's bundled Realtime session for a cascaded design. If that migration happens, evaluating Deepgram and Rime/Sarvam together avoids re-architecting the call pipeline twice.

4. Phased Rollout Plan

PHASE 1

Swap gpt-realtime-mini → gpt-realtime

Zero architecture risk — one line change in both WebSocket connect points. Re-run the same test-call battery and check whether rule #6 and stage progression hold up.

PHASE 2	**Evaluate cascaded STT + TTS pipeline** Prototype Deepgram Nova-3 (STT) + Rime/Sarvam (TTS) alongside the current bundled pipeline. Measure real added latency and barge-in/interruption handling before committing — this is a genuine rebuild, not a config change.
PHASE 3	**Migrate if Phase 2 numbers hold** Only move production traffic once latency and interruption handling in the cascaded pipeline matches or beats customer tolerance observed today.

5. The Decision This Really Comes Down To

Stay bundled (OpenAI Realtime only): keeps today's sub-500ms latency and simple single-WebSocket architecture, but permanently caps voice authenticity (no Indian accent) and transcription robustness (hallucination-on-silence stays, regardless of which OpenAI STT variant is selected).

Go cascaded (separate STT / LLM / TTS vendors): unlocks Indian-accented voices and confidence-aware transcription, at the cost of materially higher latency and the engineering work of handling turn-taking, barge-in, and interruption yourself — work the Realtime API currently does for you.

Phase 1 is free information: it tells you how much of the “it doesn't sound right” feedback is a reasoning problem (fixable today) versus a voice/accent problem (only fixable by committing to the rebuild).

Evidence sourced from backend/plivo_call.log and live MongoDB call records, 26–27 October 2026 test calls.